Methods in Short-Horizon Alpha Research: From Statistical Arbitrage to Microstructure Signals

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Abstract

This article surveys the principal methods used in short-horizon alpha research for systematic trading, with emphasis on cryptocurrency perpetual futures. We cover five broad families of approach: (1) mean-reversion and statistical arbitrage via cointegration and PCA residualization, (2) microstructure signal extraction from trade-and-quote data, (3) information-theoretic bar construction, (4) signal combination and portfolio-level stacking, and (5) cost-aware backtesting and the economics of edge. Each section states the core idea, the mathematical formulation, the practical considerations, and the key references. The presentation is grounded in concrete examples drawn from BTCUSDT perpetual futures on Binance.

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1 Introduction

Alpha research is the process of identifying predictable structure in asset returns that can be translated into a profitable trading strategy. At short horizons—seconds to tens of minutes—the relevant structure is primarily *microstructural*: patterns in order flow, quote dynamics, and cross-asset co-movements that arise from the mechanics of how markets clear, rather than from macroeconomic fundamentals.

The central challenge at short horizons is that per-trade edge is small—typically 0.1–5 basis points (bps) per round-trip—and must be weighed against transaction costs (exchange fees, spread crossing, market impact) that can easily exceed the edge itself. Successful short-horizon trading therefore requires not only signal discovery but also careful cost modelling, signal stacking for diversification, and execution optimization.

This article surveys the principal methods in the field, organized by the stage of the research pipeline in which they appear.

2 Mean-Reversion and Statistical Arbitrage

2.1 The core idea

Mean-reversion alpha rests on the observation that certain price processes—or, more precisely, certain *functions* of prices—tend to revert toward a long-run mean. When the process deviates above its mean, the expected next move is downward (and vice versa). The alpha is the ability to predict the *sign* of the next move from the current deviation.

2.2 Pairs trading and cointegration

The simplest setting is two co-moving assets X_t and Y_t whose prices are individually non-stationary (random walks) but whose linear combination

$$s_t = Y_t - \beta X_t \tag{1}$$

is stationary. This is the definition of *cointegration* (Engle & Granger, 1987). The stationary spread s_t mean-reverts, and the standard signal is its z-score:

$$z_t = \frac{s_t - \mu_s}{\sigma_s} \tag{2}$$

where μ_s and σ_s are rolling estimates of the spread's mean and standard deviation. The trading rule is: go long the spread when $z_t \ll 0$ (spread is cheap), go short when $z_t \gg 0$.

Practical considerations. The hedge ratio β is not constant. In practice it is estimated via rolling OLS, or—more elegantly—via a **Kalman filter** (Hamilton, 1994), which allows β_t to evolve as a latent state. The Kalman filter simultaneously estimates the current spread and the uncertainty around β_t, providing a natural measure of regime stability.

2.3 PCA residualization (Avellaneda–Lee)

Pairs trading is the two-asset special case of a more general framework: **statistical arbitrage via factor residualization** (Avellaneda & Lee, 2010). Given a panel of N asset returns R_{it}:

1. Perform PCA on the standardized return matrix. The first K principal components capture the common factors (market beta, sector, etc.).

2. For each asset i, regress its returns on the retained PCs:

$$R_{it} = \sum_{k=1}^{K} \beta_{ik} \, \mathrm{PC}_{kt} + \varepsilon_{it} \tag{3}$$

3. The residual ε_{it} is the "idiosyncratic" return, stripped of common factor exposure.
4. If ε_{it} is mean-reverting (negative lag-1 autocorrelation), trade the z-score of its cumulative sum (the "s-score" in Avellaneda–Lee terminology).

This construction generalizes pairs trading in two ways: it handles $N > 2$ assets simultaneously, and it strips *all* dominant factors (not just a single co-movement) before testing for mean-reversion.

Application to crypto. On a panel of BTC/ETH/SOL USDT perpetual futures, PC1 typically explains ∼85% of joint variance and is interpretable as the common "crypto market" factor. After stripping PC1, the BTC residual exhibits lag-1 autocorrelation of $\rho_1 \approx -0.11$ to -0.13—significantly more negative than the raw BTC return ($\rho_1 \approx 0$)—confirming that the idiosyncratic component is more mean-reverting than the total return.

2.4 Ornstein–Uhlenbeck modelling

The continuous-time analog of a discrete mean-reverting process is the **Ornstein–Uhlenbeck (OU) process**:

$$ds_t = \kappa \, (\theta - s_t) \, dt + \sigma \, dW_t \tag{4}$$

where $\kappa > 0$ is the rate of mean-reversion, θ is the long-run mean, and σ is the diffusion coefficient. The OU framework provides:

- A **half-life** of mean-reversion: $t_{1/2} = \ln(2)/\kappa$.
- An **optimal entry/exit threshold** via dynamic programming (Leung & Li, 2015).
- A natural connection to the Kalman filter: the OU process is a continuous-time AR(1), and the Kalman filter is the optimal recursive estimator for its parameters.

3 Microstructure Signal Extraction

At horizons below ∼1 minute, the dominant source of predictability shifts from price-level patterns to **trade-and-quote dynamics**. The key objects are the *trade tape* (a timestamped sequence of prices, quantities, and aggressor sides) and the *order book* (the standing bid and ask queues).

3.1 Trade-sign autocorrelation

The aggressor sign of trade t is $\epsilon_t \in \{-1, +1\}$ (seller-initiated or buyer-initiated). In equity markets, the autocorrelation function of ϵ_t in *trade time* typically exhibits a slow, power-law decay:

$$\rho_k \sim k^{-\gamma}, \qquad \gamma \in [0.5, 1.0] \tag{5}$$

This long-memory signature reflects the prevalence of **order-splitting**: large (institutional) orders are broken into many child orders executed over time, creating persistent signed flow (Lillo et al., 2004; Bouchaud et al., 2009).

Empirical finding on crypto perps. On 7 days of BTCUSDT aggTrades, we observe $\rho_1 = +0.63$ with a fitted decay $\gamma = 0.78$—squarely within the equity-literature range. ETHUSDT shows $\rho_1 = +0.50$, $\gamma = 0.99$. SOLUSDT, being less liquid (271 ms mean inter-arrival vs. 66 ms for BTC), shows the opposite pattern at lag 1: $\rho_1 = -0.16$, the **bid–ask-bounce** signature. Two distinct microstructure regimes are cleanly separated by liquidity.

3.2 Order flow imbalance (OFI)

Aggregating signed trade volume over a short window gives the **order flow imbalance**:

$$\mathrm{OFI}_t = \sum_{i \in \text{window}} \epsilon_i \cdot q_i \tag{6}$$

where q_i is the quantity of trade i. OFI is a measure of net buying or selling pressure. Its predictive content for future returns has been documented extensively (Cont et al., 2014; Cartea et al., 2015).

In a mean-reversion context, OFI is used as a **contrarian** signal: excess buying pressure (OFI $\gg$ 0) is expected to reverse as the transient demand subsides.

3.3 Price impact

The **price-impact function** measures how much the mid-price moves in response to a signed trade of size q:

$$\Delta p \approx \lambda \cdot f(q) \tag{7}$$

where λ is Kyle's lambda (Kyle, 1985) and $f(q)$ is typically concave (square-root law for large q):

$$f(q) \approx \sigma \sqrt{\frac{q}{V}} \tag{8}$$

with V the average daily volume (Bouchaud et al., 2009; Almgren et al., 2005).

Understanding the impact function is essential for **cost modelling**: any alpha signal that requires market orders will lose a portion of its edge to impact, and the loss is a function of order size. This is why the per-turn edge ($\sim$0.1–0.25 bps for a typical mean-reversion signal on BTC) must be compared against the *total* cost including impact, not just exchange fees.

3.4 Trade-clock vs. wall-clock sampling

A recurring theme in microstructure research is that financial time flows unevenly. Clark (1973) showed that price variance per unit of calendar time is not constant, but scales with the number of trades ("subordinated stochastic process"). The practical implication: signals computed on **trade-time** or **volume-time** bars capture more of the predictable structure than signals on fixed wall-clock bars.

AR(1) comparison. On BTCUSDT, signed-flow AR(1) R^2 peaks at $\sim$0.023 in 5-second wall-clock buckets but is $\sim$0.63 at lag-1 in trade-time. The difference is striking: aggregating into wall-clock buckets averages bursts with quiet periods and destroys most of the signal.

3.5 Cross-symbol lead–lag

In multi-asset settings, one asset's order flow may predict another asset's price movement with a short delay. The standard diagnostic is the **cross-correlation of signed flow** at sub-second lags.

In crypto markets, the conventional belief is that BTC flow "leads" altcoin flow. Empirically, this is not always detectable at 10–100 ms resolution on the same exchange. The lead may exist at sub-10 ms timescales (relevant for HFT) or in different variables (price rather than flow).

4 Information-Theoretic Bar Construction

4.1 Dollar bars and volume bars

de Prado (2018) argues that fixed-time bars (e.g., 1-minute OHLCV) are a poor sampling scheme for financial data because they treat all minutes as containing equal information. Instead, he proposes **information-driven bars**:

- **Volume bars**: close a bar every V units of base currency traded.
- **Dollar bars**: close a bar every $\$D$ of notional traded.
- **Tick bars**: close a bar every N trades.

The motivation: by equalizing the "information content" per bar, these schemes produce returns that are closer to i.i.d. normal, which makes statistical tests (autocorrelation, z-scores) more reliable.

4.2 Empirical test

On 7 days of BTCUSDT aggTrades, \$5M dollar bars produce a BTC residual with lag-1 autocorrelation $\rho_1 = -0.13$ (vs. -0.11 on 1-minute time bars). The signal concentrates slightly, validating the hypothesis, but the per-turn edge remains bounded by $|\rho_1|\,\sigma^2 \approx 0.1$ bps—a ceiling imposed by the horizon, not the bar construction.

5 Signal Combination and Stacking

5.1 The fundamental law of active management

Grinold & Kahn (2000) formalize the relationship between signal quality and portfolio performance:

$$\mathrm{IR} \approx \mathrm{IC} \times \sqrt{\mathrm{BR}} \tag{9}$$

where IR is the information ratio (annualized Sharpe of the active return), IC is the information coefficient (correlation between signal and realized return), and BR is the *breadth* (number of independent bets per year).

The practical implication: a mediocre signal (IC = 0.02) applied to 50 instruments with daily rebalancing (BR = 50 × 252 = 12,600) achieves IR $\approx 0.02 \times \sqrt{12{,}600} \approx 2.2$—a perfectly viable strategy despite the tiny per-bet IC.

5.2 Stacking uncorrelated signals

Given N signals with equal Sharpe S and average pairwise correlation $\bar{\rho}$, the portfolio Sharpe is approximately:

$$S_{\text{port}} \approx \frac{\sqrt{N}}{\sqrt{1 + (N-1)\bar{\rho}}} \cdot S \tag{10}$$

When $\bar{\rho} \approx 0$, the improvement is $\sqrt{N}$—i.e., 15 uncorrelated signals of Sharpe 2 produce a portfolio of Sharpe 8. When $\bar{\rho}$ is large (> 0.5), the improvement saturates quickly and adding more signals of the same type provides diminishing returns.

Empirical demonstration. On 12 months of BTCUSDT 1-minute klines, three signal families (composite mean-reversion, funding-rate carry, PCA residual) have average pairwise correlation $\bar{\rho} = 0.08$. The equal-weight stack achieves a realized Sharpe multiplier of 1.71×, close to the theoretical $\sqrt{3} \approx 1.73\times$.

5.3 Weighting schemes

Common approaches to signal weighting:

1. **Equal weight**: $w_i = 1/N$. Simple, robust, hard to beat out of sample.
2. **Inverse volatility**: $w_i \propto 1/\sigma_i$ where σ_i is the volatility of signal i's PnL. A risk-parity analog for signals.
3. **Markowitz (mean-variance)**: $\mathbf{w} \propto \Sigma^{-1}\boldsymbol{\mu}$. Theoretically optimal but notoriously unstable in practice due to estimation error in Σ and $\boldsymbol{\mu}$.
4. **Shrinkage / regularized**: Ledoit–Wolf shrinkage on Σ, ridge/LASSO on $\boldsymbol{\mu}$. The practical standard at most systematic funds.

6 Cost-Aware Backtesting and the Economics of Edge

6.1 The edge-per-turn framework

The single most important metric for a short-horizon signal is **edge per turn** (in basis points):

$$\text{edge}_{\text{bps}} = \frac{\text{gross PnL}}{\text{total turnover}} \times 10{,}000 \tag{11}$$

This number is independent of bar frequency, position sizing, and leverage—it measures the raw informational content of the signal per unit of trading activity.

A signal is viable if and only if:

$$\text{edge}_{\text{bps}} > \text{all-in cost per turn (bps)} \tag{12}$$

where all-in cost includes:

- Exchange fee (taker: 1.7–4.5 bps; maker: 0–0.5 bps on Binance USD-M futures, depending on VIP tier)
- Half-spread (typically 0.01–0.1 bps for BTC perps)
- Market impact (a function of order size; negligible for small orders but dominant for large ones)

6.2 The $|\rho_1|\sigma^2$ bound

For a pure mean-reversion signal on a single asset at a fixed bar frequency, the expected per-bar gross PnL of a unit-sized position is approximately:

$$\mathbb{E}[\text{gross PnL per bar}] \approx |\rho_1| \cdot \sigma^2 \tag{13}$$

where ρ_1 is the lag-1 autocorrelation of returns and σ is the per-bar volatility. This bounds the achievable edge: at 1-minute bars on BTC ($\sigma \approx 7 \times 10^{-5}$, $|\rho_1| \approx 0.1$), the product is ∼0.1 bps—an order of magnitude below taker cost.

This bound motivates three paths to profitability:

1. **Maker execution**: flip the cost sign (collect spread instead of paying it).
2. **Shorter horizon**: push to sub-second bars where $|\rho_1|$ is larger (e.g., sign ACF $\rho_1 = +0.63$ in trade time) even though σ is smaller—the product can be 10–100× the 1-minute value.
3. **Cross-sectional breadth**: apply the same signal to 50+ instruments, reducing per-name turnover and gaining $\sqrt{N}$ diversification.

6.3 Walk-forward validation

A backtest on the full sample is subject to overfitting. The standard remedy is **walk-forward validation**: the sample is divided into rolling train/test windows (e.g., 3-month train / 1-month test), the signal parameters are estimated on the train window, and PnL is evaluated on the out-of-sample test window only. A strategy that is consistently positive across all folds is more credible than one with a high full-sample Sharpe but variable OOS performance (Bailey et al., 2014).

6.4 Paper trading as pipeline validation

Live paper trading—running the signal computation and simulated execution against a real-time data feed—serves as a final check that:

1. The backtest implementation is correct (no look-ahead bias, no survivorship bias, no stale-data artifacts).
2. The live signal distribution matches the historical one (no regime shift since the backtest period).
3. The execution model's assumptions (fill at mid ± half-spread, no queue priority, no partial fills) produce PnL consistent with the backtest prediction.

Paper trading does not validate profitability—it validates the *pipeline*. If live PnL diverges sharply from the backtest prediction, the backtest is suspect regardless of its Sharpe.

7 Advanced Methods

7.1 Kalman filter for time-varying hedge ratios

In pairs trading, the hedge ratio β drifts over time. The **Kalman filter** models this as a linear-Gaussian state-space system:

$$\text{State:} \quad \beta_t = \beta_{t-1} + w_t, \quad w_t \sim N(0, Q) \tag{14}$$

$$\text{Observation:} \quad Y_t = \beta_t X_t + v_t, \quad v_t \sim N(0, R) \tag{15}$$

The filter recursively updates $\hat{\beta}_t$ and its uncertainty P_t at each new observation. The residual $v_t = Y_t - \hat{\beta}_t X_t$ is the spread, and its z-score (scaled by $\sqrt{P_t + R}$) is the trading signal.

This is more adaptive than rolling OLS and naturally down-weights periods of high parameter uncertainty, which makes the signal more conservative during regime transitions.

7.2 Hidden Markov Models for regime detection

Markets alternate between regimes (trending, mean-reverting, volatile, quiet). A **Hidden Markov Model (HMM)** with K states provides a probabilistic estimate of the current regime:

$$p(S_t = k \mid \text{data}) = \text{filtered probability} \tag{16}$$

The HMM parameters (transition matrix, per-state return distribution) are estimated via the Baum–Welch algorithm. The signal is then *conditioned* on the regime: trade the mean-reversion signal only when $p(\text{MR regime}) > \text{threshold}$.

Practical note. HMMs are powerful but fragile: with $K > 3$ states they tend to overfit, and the Baum–Welch algorithm is sensitive to initialization. Practitioners typically use $K = 2$ ("trending" vs. "mean-reverting") and re-estimate parameters monthly.

7.3 Hawkes processes for trade arrival modelling

Trade arrivals on crypto exchanges are bursty: clusters of rapid-fire trades are separated by quiet intervals. The **Hawkes process** models this self-exciting behavior:

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha \, e^{-\beta(t - t_i)} \tag{17}$$

where $\lambda(t)$ is the intensity (arrival rate), μ is the baseline rate, and α, β control the magnitude and decay of self-excitation.

Hawkes models are used for:

- Detecting periods of abnormal activity (potential signals or risk events).
- Modelling the impact of one's own trades on subsequent market activity ("market footprint").
- Calibrating execution algorithms that need to predict near-future trade rates.

7.4 Machine learning approaches

Modern alpha research increasingly uses ML for feature selection and non-linear signal combination:

- **LASSO / Elastic Net**: sparse linear combination of a large feature set; useful for selecting which microstructure features matter.
- **Gradient-boosted trees (XGBoost, LightGBM)**: capture non-linear interactions between features without explicit specification; risk of overfitting, mitigated by walk-forward validation.
- **Neural networks**: sequence models (LSTM, Transformer) for temporal dependencies; attention mechanisms for cross-asset structure. State of the art in some contexts, but data-hungry and hard to interpret.
- **Meta-labeling** (de Prado, 2018): a secondary model that predicts whether a primary signal's trade will be profitable, effectively learning a regime-conditional filter.

8 Conclusion

Short-horizon alpha research is not a single technique but a *pipeline*: from raw tick data, through bar construction and signal extraction, to cost-aware backtesting and portfolio-level stacking. The methods span classical time-series econometrics (cointegration, PCA, AR models), market microstructure theory (Kyle's lambda, Lo–MacKinlay sign ACF, Hasbrouck's information share), modern portfolio theory (Grinold–Kahn, risk parity), and machine learning (LASSO, gradient boosting, sequence models).

The unifying insight is that no single signal at the short horizon carries enough edge to overcome transaction costs alone. The business of systematic short-horizon trading is therefore a *portfolio construction* problem: the researcher's job is to produce signals that are individually non-zero and mutually uncorrelated, so that $\sqrt{N}$ does the heavy lifting at the portfolio level.

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